

Cloud Computing

Semester 2026-III

Course Project Description

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Overview

The course project of *Cloud Computing* is a semester-long, team-based systems design and implementation project built around the challenge of designing, architecting, and validating a cloud-native distributed system for a real-world, high-traffic application. Each team has been assigned a unique application — such as Uber, DoorDash, Netflix, Airbnb, Stripe, or Spotify — and must design a complete, production-grade cloud architecture capable of handling the application’s real-world workload characteristics.

The project follows a **Four-Delivery Methodology** that mirrors the professional progression of a cloud systems architect:

- i. **Delivery 1 — Requirements & Workload Model:** Understand the business problem and translate it into functional and non-functional requirements, culminating in a formal workload model.
- ii. **Delivery 2 — Software Architecture:** Design the software architecture independently of any cloud provider, defining services, data architecture, communication patterns, and architecture decisions.
- iii. **Delivery 3 — Cloud Architecture & Scalability:** Translate the software architecture into a concrete cloud deployment plan, including compute, networking, data, operations, security, cost analysis, and a $20\times$ growth scenario.
- iv. **Delivery 4 — Prototype, Resilience & Final Architecture:** Validate the architecture using an **AWS Academy** vertical slice prototype, analyze resilience under failure scenarios, and produce a final integrated architecture document.

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The conceptual progression of the four deliveries is deliberately structured so that the cloud provider does not appear until Delivery 3. Students first learn to think as software architects, then as infrastructure architects, and finally use **AWS Academy** to validate their design decisions with real AWS services.

Project Topic

Each team works on a different real-world high-traffic application, assigned at the beginning of the semester. The assigned applications are:

- **Team 1 — Uber:** Real-time ride-sharing platform with geolocation matching, surge pricing, and payment coordination
- **Team 2 — DoorDash:** Food delivery platform with order management, route optimization, and merchant integration
- **Team 3 — Netflix:** Video streaming platform with content delivery, personalization, and recommendation engine
- **Team 4 — Airbnb:** Short-term rental marketplace with booking engine, calendar management, and search
- **Team 5 — Stripe:** Payment processing platform with transaction management, fraud detection, and settlement
- **Team 6 — Spotify:** Music streaming platform with personalization, collaborative filtering, and media delivery

Each application exhibits unique high-traffic patterns, consistency requirements, and architectural challenges that make it an ideal subject for cloud-native distributed systems design.

Project Scope and Objectives

- **Systems Thinking:** Develop the ability to reason about complex distributed systems from business requirements all the way to cloud infrastructure decisions.
- **Architecture-First Approach:** Practice designing software and cloud architectures before writing any code, ensuring that implementation decisions are grounded in requirements and trade-off analysis.
- **Cloud-Native Design:** Apply cloud-native principles including stateless services, auto-scaling, managed databases, event-driven communication, and infrastructure as code.
- **Quantitative Reasoning:** Support all architecture decisions with quantitative justification: workload calculations, instance sizing, cost estimates, and capacity planning under growth scenarios.

- **Resilience Engineering:** Analyze how the proposed architecture behaves under realistic failure scenarios and define detection, failover, and recovery strategies.
- **Prototype Validation:** Use AWS Academy to implement a minimum vertical slice that demonstrates the technical feasibility of the proposed cloud architecture.
- **Professional Documentation:** Produce architecture documents, diagrams, and reports that meet professional engineering standards.

Four-Delivery Summary

1. Delivery 1 — Requirements & Workload Model:

- System description: business problem, actors, use cases, system boundaries
- Functional requirements (8–12 requirements)
- Non-functional requirements: availability, response time, data volume, geographic scope
- Formal workload model: average and peak requests/hour, read/write ratio, data growth
- *No cloud provider selection at this stage*

2. Delivery 2 — Software Architecture:

- System context diagram
- Service-oriented software architecture with defined service boundaries
- Service definitions: responsibility, APIs, data owned, dependencies, scaling characteristics
- Data architecture: relational vs. NoSQL, ownership, consistency, transactions
- Communication architecture: synchronous and asynchronous interactions with justification
- Architecture Decision Records (ADRs) for important design decisions
- *Completely cloud-independent design*

3. Delivery 3 — Cloud Architecture & Scalability:

- Cloud architecture diagram translating software services to cloud infrastructure
- Compute: containers, instance types, auto-scaling policies, stateless/stateful considerations
- Networking: VPC design, public/private subnets, load balancing, security groups
- Data: managed databases, object storage, caching, backup strategies
- Operations: logging, monitoring, metrics, health checks
- Security: authentication, authorization, secrets management, network security

- Growth scenario: architecture evolution for 20× workload increase
- Cost analysis: monthly estimate per component category

4. Delivery 4 — Prototype, Resilience & Final Architecture:

- AWS Academy vertical slice: API Gateway → Service → Database + Queue → Worker
- Resilience analysis: instance failure, AZ failure, database failure scenarios
- Final integrated architecture document incorporating all lessons learned
- Performance analysis: normal load, peak load, 20× growth
- Final cost breakdown

Deliverables Summary

Each team will produce five major deliverables throughout the semester:

- **Requirements & Workload Specification (Delivery 1)** — Business context, requirements catalog, and quantitative workload model
- **Software Architecture Document (Delivery 2)** — Cloud-independent architecture with services, data, communication, and ADRs
- **Cloud Architecture & Capacity Plan (Delivery 3)** — Infrastructure diagram, capacity calculations, growth scenario, and cost estimate
- **Resilience Analysis Report (Delivery 4)** — Failure scenario analysis with detection, failover, recovery, and impact assessment
- **Final Integrated Project (Final Delivery)** — AWS Academy prototype, revised architecture, and comprehensive final report

Submission Requirements

- Each delivery must be submitted as a PDF report through the designated course platform by the specified deadline
- Each delivery PDF must follow the naming convention: `groupN_deliveryM.pdf`, where N is the team number and M is the delivery number (e.g., `group1_delivery1.pdf`, `group3_delivery2.pdf`)
- All source code, AWS configuration, and infrastructure-as-code must be maintained in a GitHub repository with a clear structure
- All documentation must be in **English** and follow professional technical writing standards

- Architecture diagrams must use standard notation (C4, UML, or AWS/cloud architecture diagram conventions) and be included in the report
- Cost estimates must cite official cloud provider pricing pages as of the submission date
- The AWS Academy prototype in Delivery 4 must be reproducible from the repository following only the README.md instructions

This course project is the intellectual core of your Cloud Computing journey. Through it, you will practice the full professional arc of a cloud architect: from listening to a business problem, to designing a system, to deploying it on cloud infrastructure, to validating that it holds under real-world pressure. These are exactly the skills that define a modern cloud engineer. Good luck, and design boldly!